

ml-1.py - C:/Users/chait/AppData/Local/Programs/Python/Python313/ml-1.py (3.13.5)

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```
import heapq

graph = {
    'A': [('B', 2), ('C', 4)],
    'B': [('D', 3), ('E', 1)],
    'C': [('F', 5)],
    'D': [],
    'E': [('F', 2)],
    'F': []
}

queue = [(0, 'A')]
visited = set()

while queue:
    cost, node = heapq.heappop(queue)

    if node not in visited:
        print(node, "Cost =", cost)
        visited.add(node)

        for neighbor, weight in graph[node]:
            heapq.heappush(queue, (cost + weight, neighbor))
```

IDLE Shell 3.13.5

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Python 3.13.5 (tags/v3.13.5:6cb20a2, Jun 11 2025, 16:15:46) [MSC v.1943 64 bit (AMD64)] on win32
Enter "help" below or click "Help" above for more information.

```
>>>
===== RESTART: C:/Users/chait
/AppData/Local/Programs/Python/Python313/ml-1.py =====
A B C D E F
>>>
===== RESTART: C:/Users/chait/AppData/Local/Programs/Python/Python313/ml-2.py =====
A B D E C F
>>>
===== RESTART: C:/Users/chait/AppData/Local/Programs/Python/Python313/ml-1.py =====
A Cost = 0
B Cost = 2
E Cost = 3
C Cost = 4
D Cost = 5
F Cost = 5
>>>
```

Ln: 17 Col: 0

Ln: 24 Col: 0